Cell imaging and machine learning to advance genetic research and accelerate drug discovery

MIT 5.08/7.08 Fundamentals of chemical biology guest lecture



Runxi Shen, Ph.D.

Postdoctoral Associate, Carpenter-Singh Lab, Imaging Platform

March 18, 2026

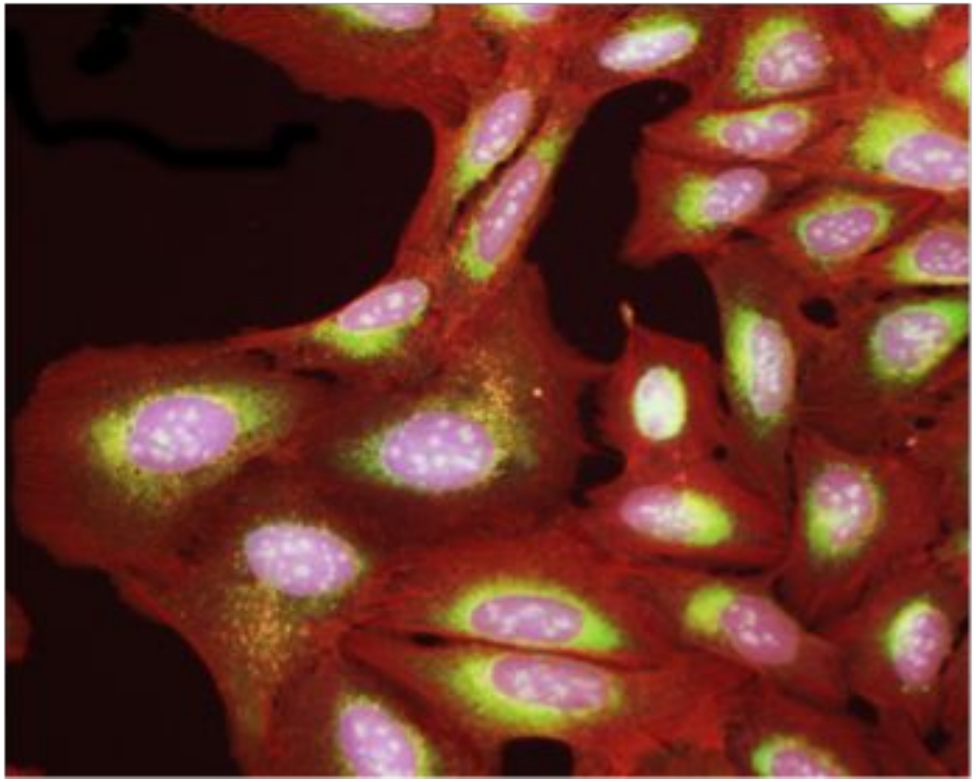
BROAD INSTITUTE

Images of cells tell us a lot of information about cell biology

Microscopy: single-cell resolution

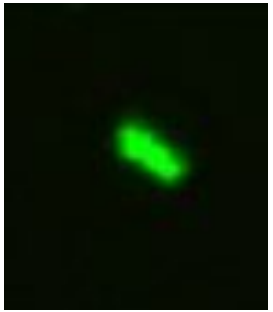
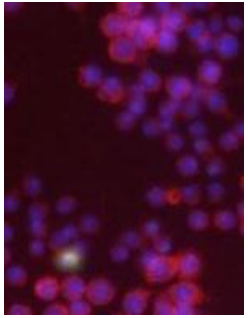
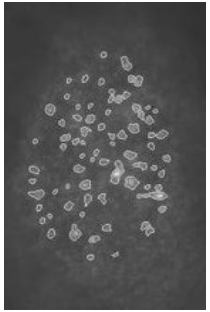
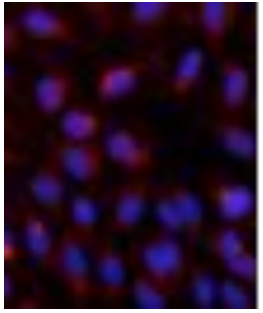
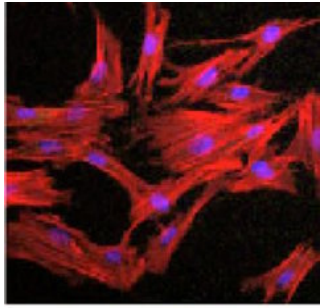
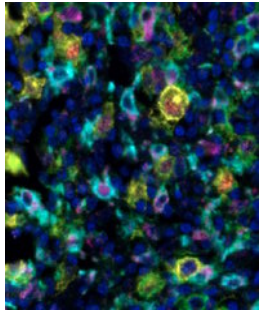
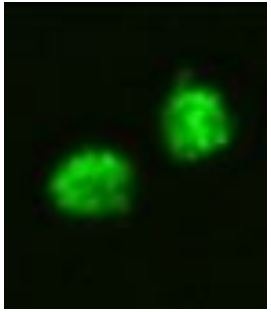
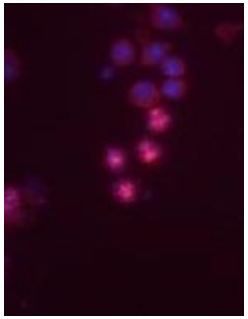
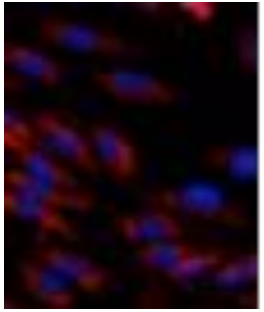
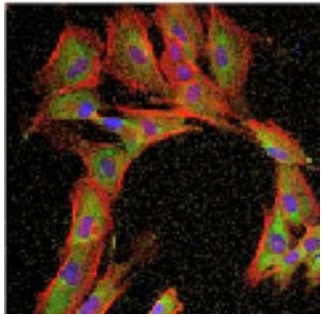
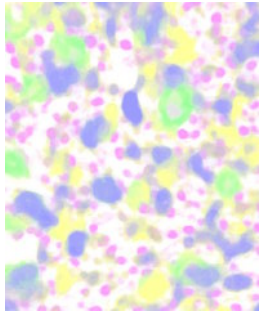
Since the 1600's

- Information about how different molecules interact
- Ability to extract information about live cells
- Ability to see not just if a molecule's expression level has changed, but if its "behavior" in the cell has changed
- Closer to reporting on cell function than genomics, transcriptomics and even proteomics





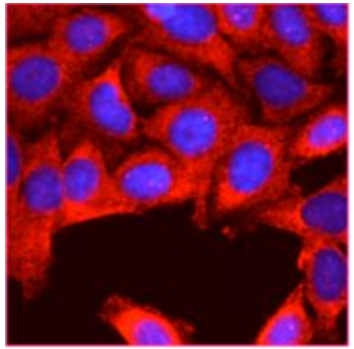
Screen all the things!

Mitosis	Meiosis	DNA damage	Mitochondrial abundance	mTOR pathway activation	Immune environment
					
					
Mitchison lab	Orr-Weaver lab	Yaffe lab	Mootha lab	Sabatini lab	Shipp/Rodig labs

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Measure ***everything*** in a single cell

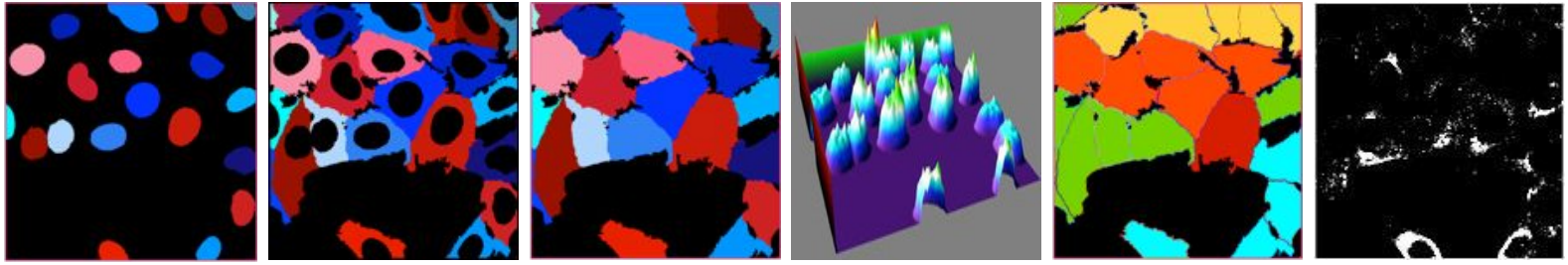






CellProfiler™
cell image analysis software

“Everything” = Counts, Shapes, Sizes, Intensities, Textures, Correlations, Relationships

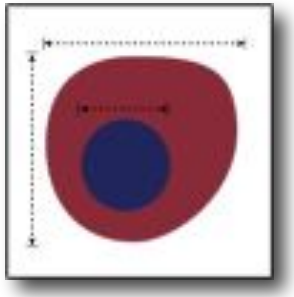


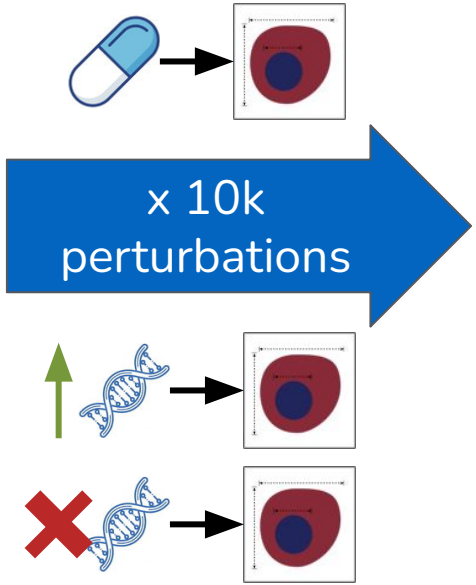
Thousands of features / cell

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A revolution in how we think about images and how we use images

Measure known phenotypes





Profile to characterize samples

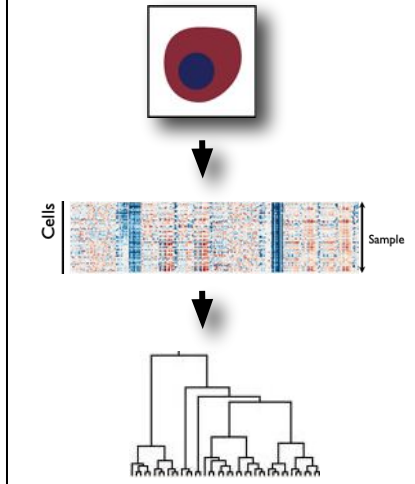
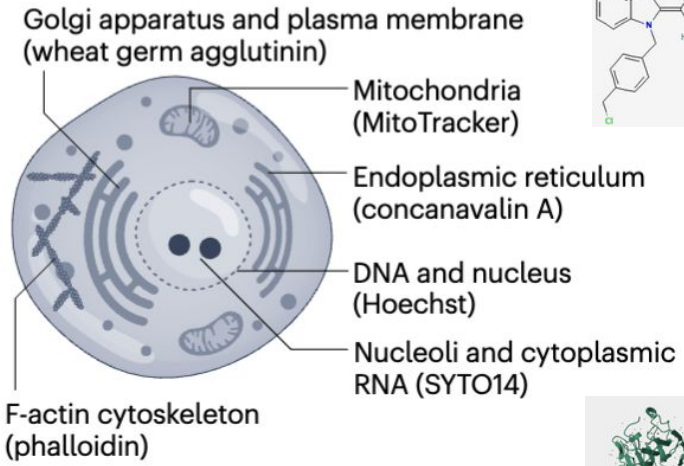


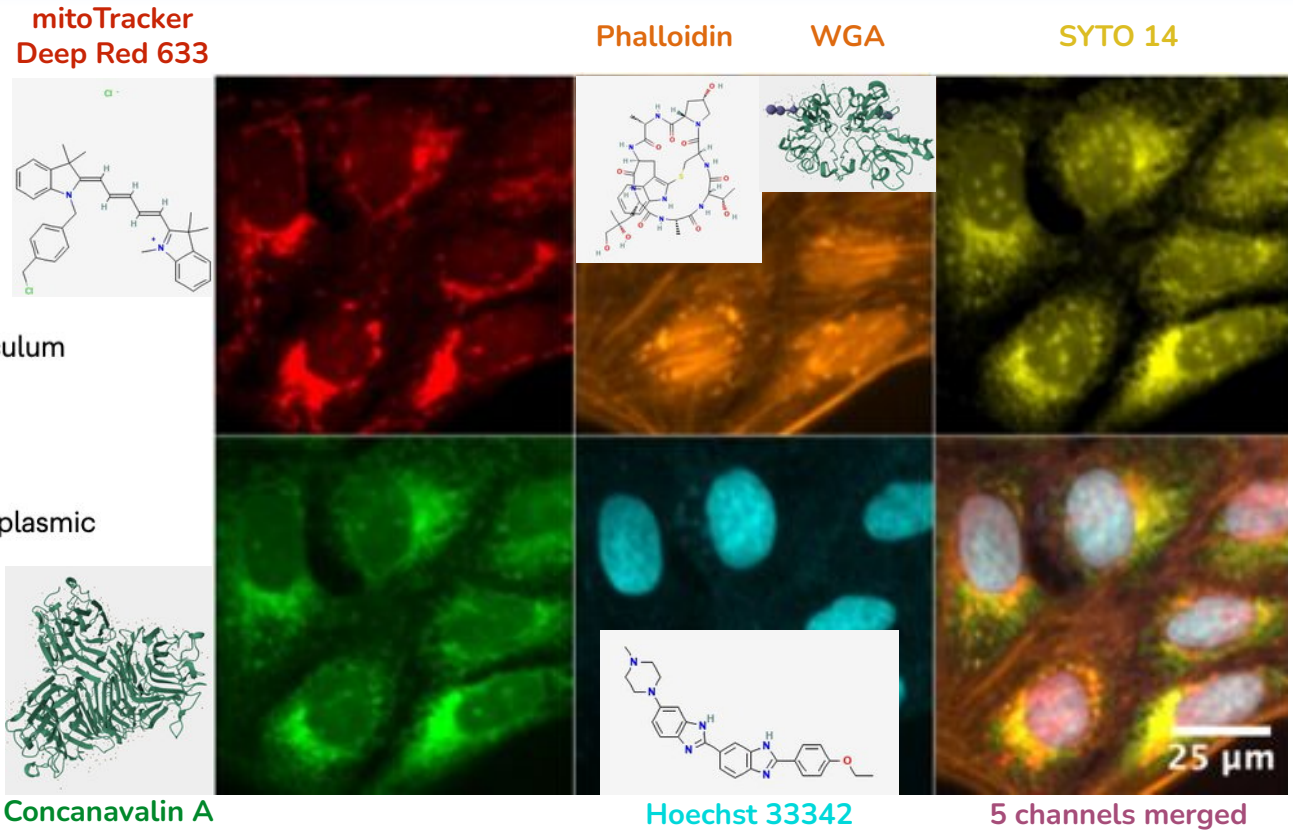
Image-based **screening**
(High-content screening)

Image-based **profiling**
(using a high-throughput assay)

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The Cell Painting assay maximizes information for image-based profiling



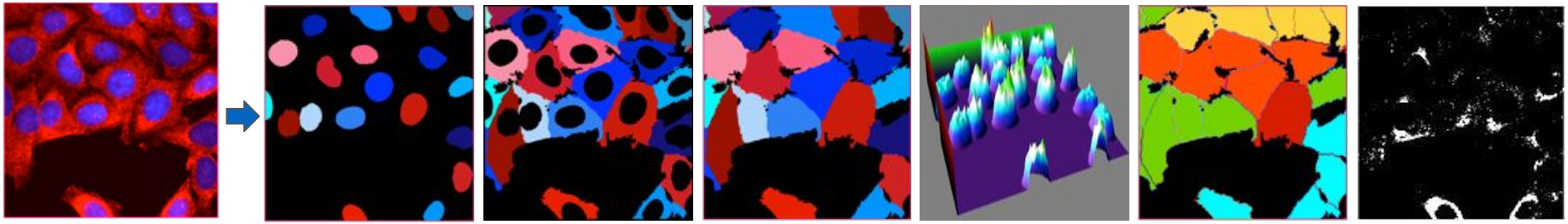


Gustafsdottir, et al. PLOS ONE 2013 | Bray, et al. Nature Protocols 2016
Cimini et al. Nature Protocols 2023 | Seal et al. Nature Methods, 2025

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INSTITUTE

Measure everything in cell images in a high-throughput way

"Everything" = Counts, Shapes, Sizes, Intensities, Textures, Correlations, Relationships



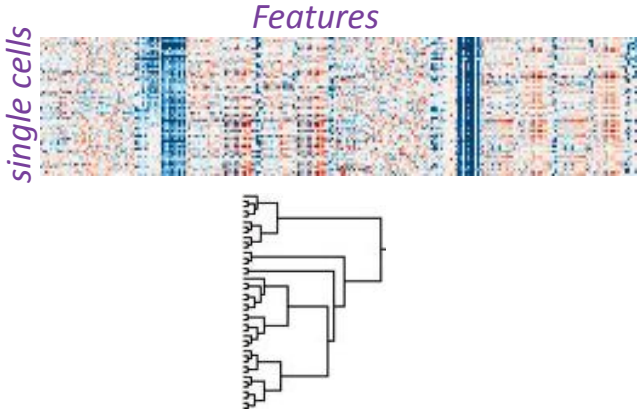
~3,000 features / cell

Sample prep



100's of 384-well plates
100's of thousands of samples

CellProfiler™
cell image analysis software



Why imaging?

Close to cell functions

10x cheaper than bulk mRNA and 1000-plex protein profiling

1000x cheaper than single-cell RNA-seq

But similar predictive success!

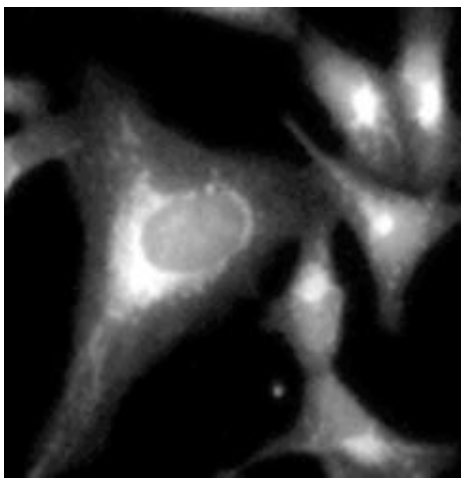
And, complementary to mRNA/protein

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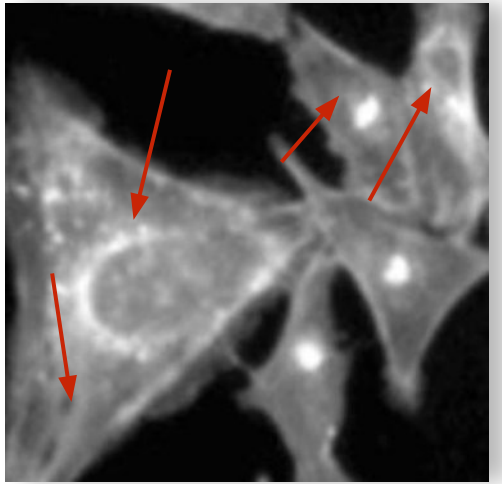
Details of the PerkinElmer Opera Phenix channels and stains imaged in the Cell Painting assay

Dye	Filter (excitation; nm)	Filter (emission; nm)	Organelle or cellular component	CellProfiler channel name
Hoechst 33342	405	435-480	Nucleus	DNA
Concanavalin A/Alexa Fluor 488 conjugate	488	500-550	Endoplasmic reticulum	ER
SYTO 14 nucleic acid stain	488	570-630 (widefield), 515-550 (confocal)	Nucleoli, cytoplasmic RNA	RNA
Phalloidin/Alexa Fluor 568 conjugate, wheat-germ agglutinin/Alexa Fluor 555 conjugate	561	570-630	F-actin cytoskeleton, Golgi, plasma membrane	AGP
MitoTracker Deep Red	640	650-760	Mitochondria	Mito

Can the image of a diseased cell tell us what's wrong with it?



Healthy



Disease?

Images do often faithfully reflect diseases



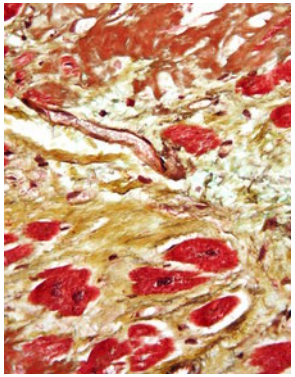
Prediction:Down's
Confidence:0.999



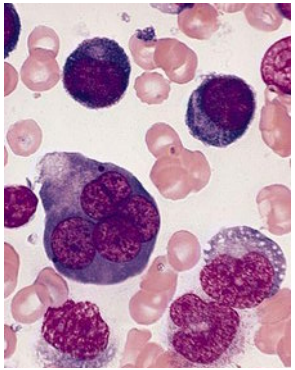
Prediction:Angelman
Confidence:0.997

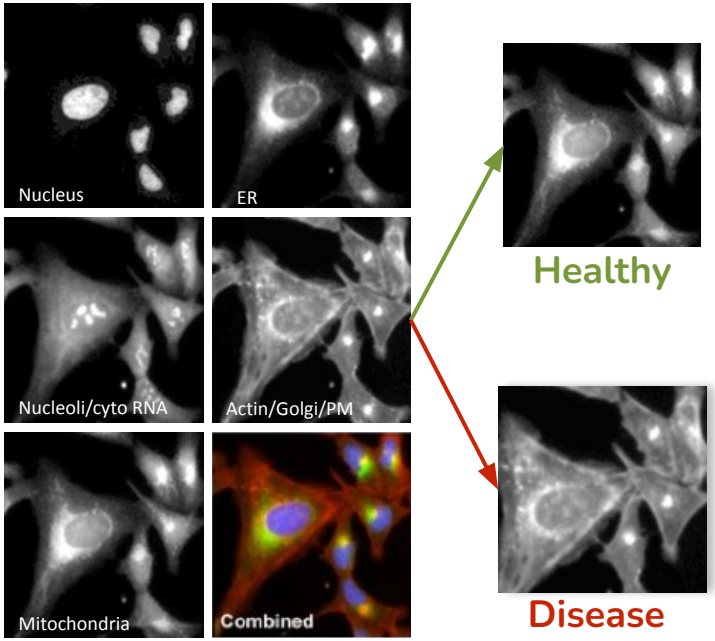
Singh & Kisku, IEEE ISED 2018 eLife
DB from Ferry, et al. 2014





Wikipedia





The Cell Painting assay

If cell images can tell us they're diseased then we could screen drugs to find the treatments

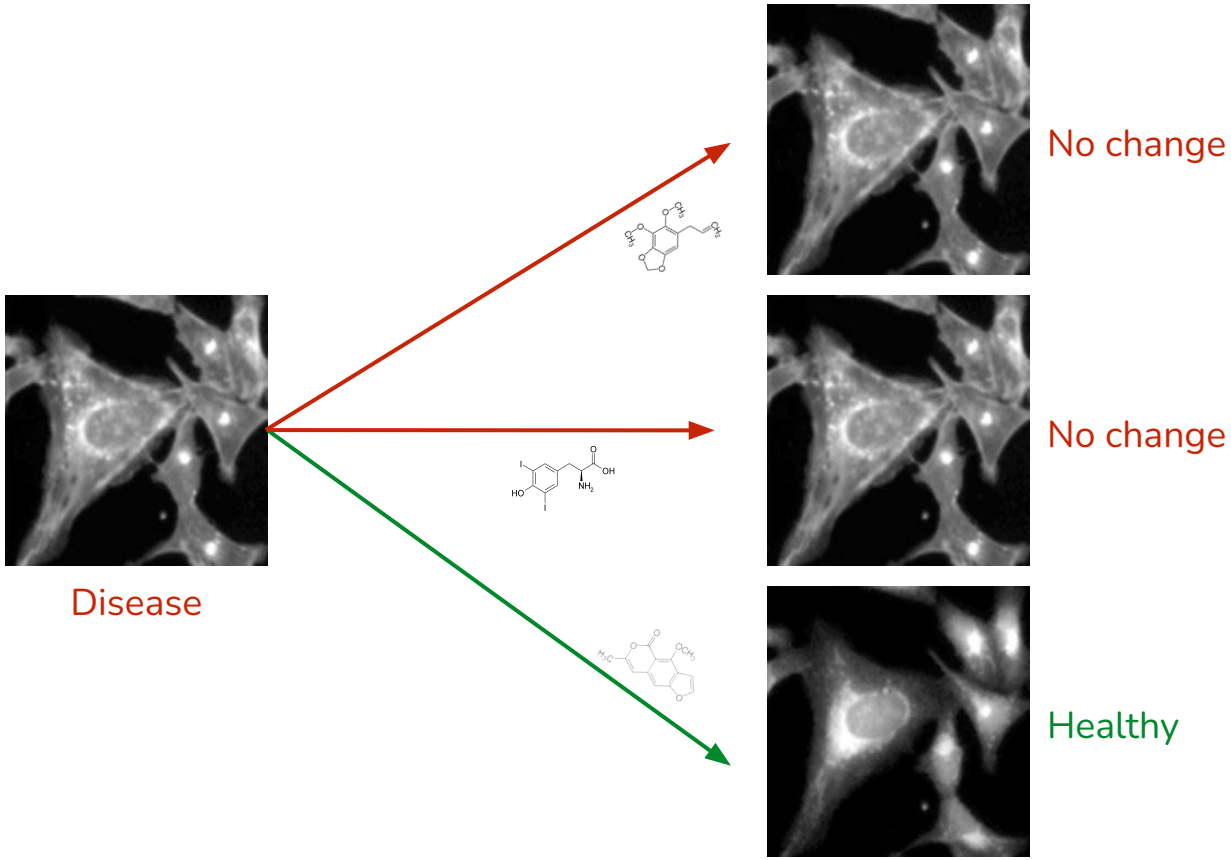
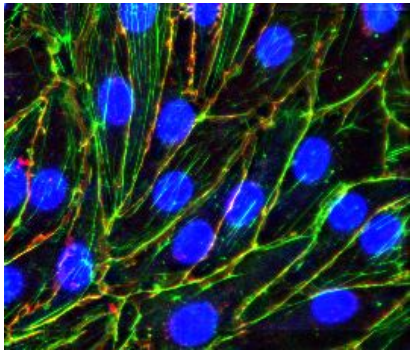


Image-based profiling to find the proper treatment for every disease phenotype by screening existing drugs

Healthy



existing + drug?

Can we find drugs that reverse every disease phenotype?

Can we find a phenotype for every disease?

(more cheaply)

Disease

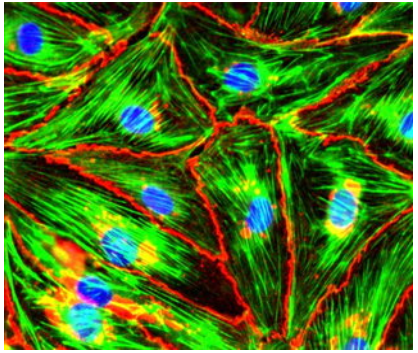
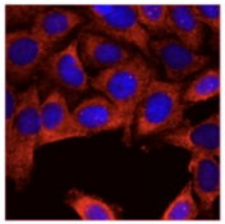
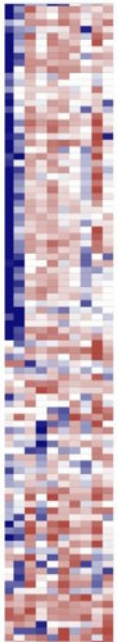


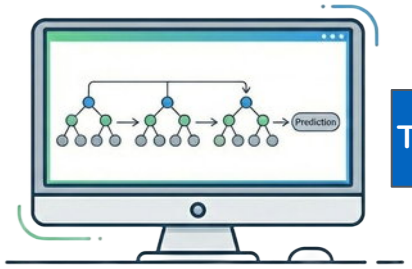
Image-based profiling and machine learning to understand genetic variants, accelerate therapeutic discovery, and predict compound toxicity



Cell images



Feature sets

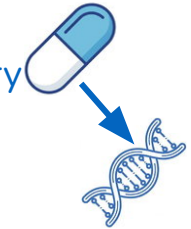


Modeling

Variants → Characterize the functional impacts of genetic variants



Treatment → Accelerate drug discovery in rare disease by drug repurposing



Drugs → Predict compound toxicity using in-vitro assay



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Rare diseases present a major unmet need



That's 10%!
(330 million worldwide)



They cause 35% of deaths in the first year of life

10,000+
rare diseases



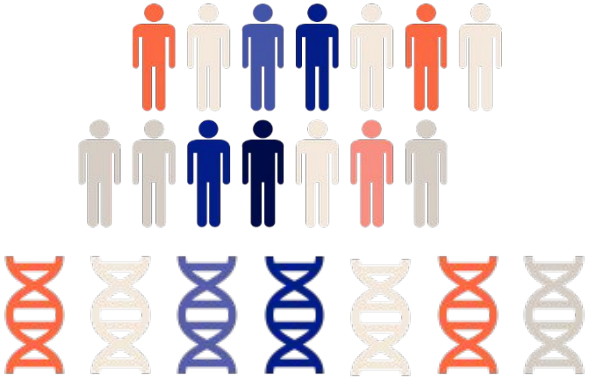
APPROVED TREATMENTS ARE AVAILABLE FOR ONLY **5%** OF ALL RARE DISEASES

Current state of the field:

- ❑ Customize a protein or cell assay for the gene's function (~6 months) OR
- ❑ Do Cell Painting on patient cell lines, then screen novel chemicals to find a potential hit (~6 months)
- ❑ **Total cost $2 million per disease to get a hit** (estimate: Bay Bridge Bio)

Many rare diseases are caused by genetic variations among the human population

Genetic variations in the human population





Prediction:Angelman

Genome Sequenced



Variants Identified

Two clinical types of variations in the human genome



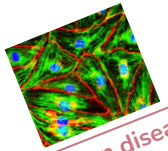
Pathogenic

Disease phenotype



Benign

Healthy phenotype



Known disease association

Unknown Way More Common!!!

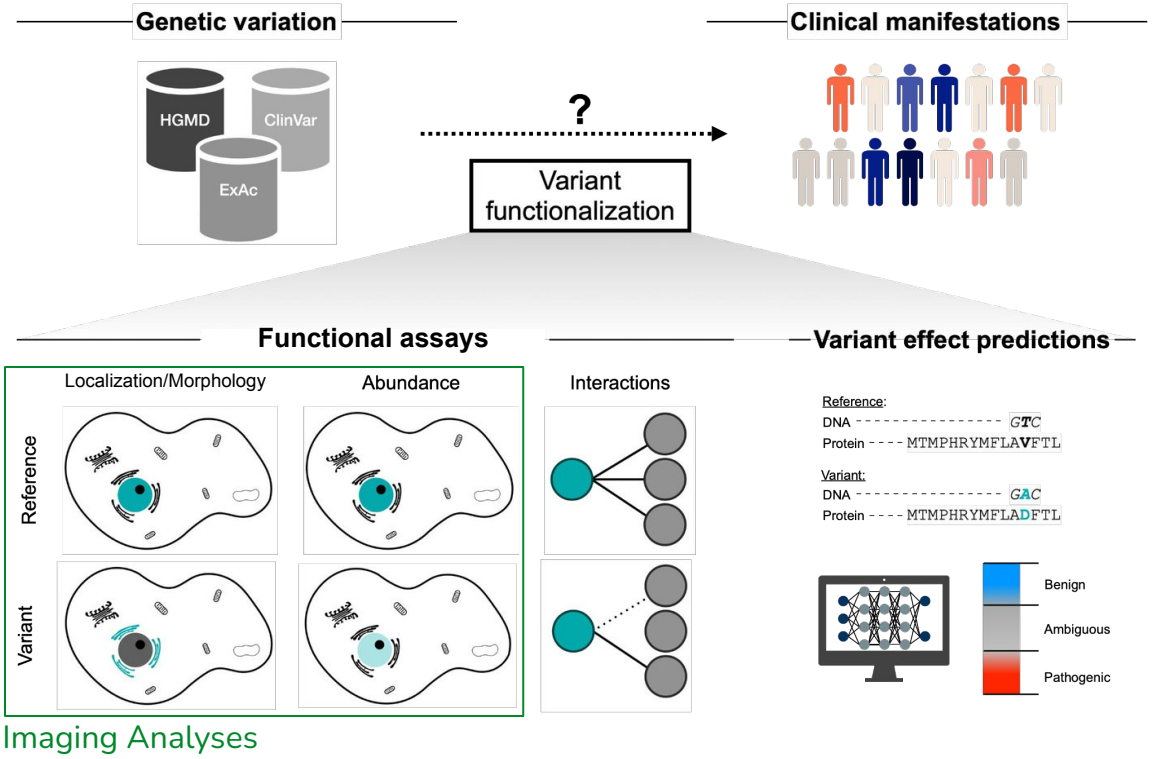
?????

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Functional assays can reveal the impacts of genetic variants

VarChAMP: Variant Characterization across the Mendelian Proteome

Clone & overexpress ~30k missense variants in >2000 rare disease genes



Imaging Analyses

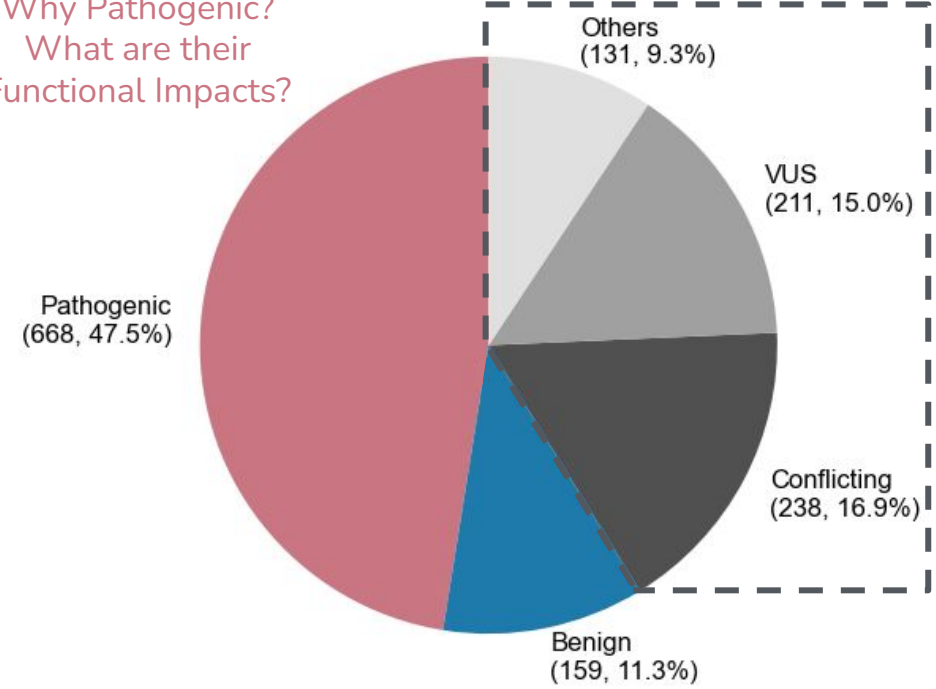
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Insights into variant function can lead to more accurate diagnosis and new treatments

ClinVar Categories of >1,400 Profiled Variants

Why Pathogenic? What are their Functional Impacts?

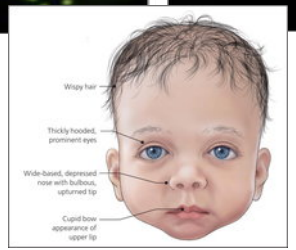
Are they pathogenic or not?



Identify the phenotypes



Noonan syndrome



Bhambhani & Muenke, Am Fam Physician. 2014

Screen drugs to reverse them



Mikko Taipale
U Toronto



Jessica Lacoste

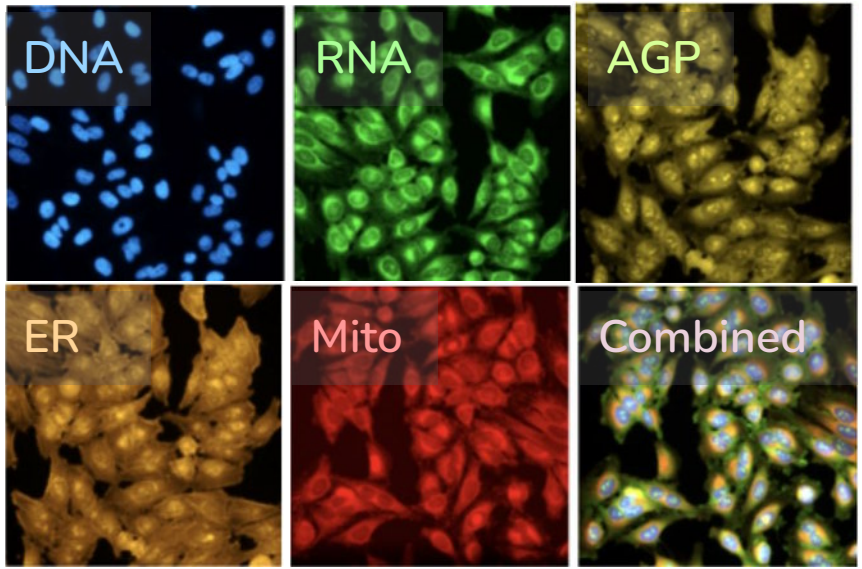


Marzieh Haghighi

Lacoste and Haghighi, et al. Cell 2024

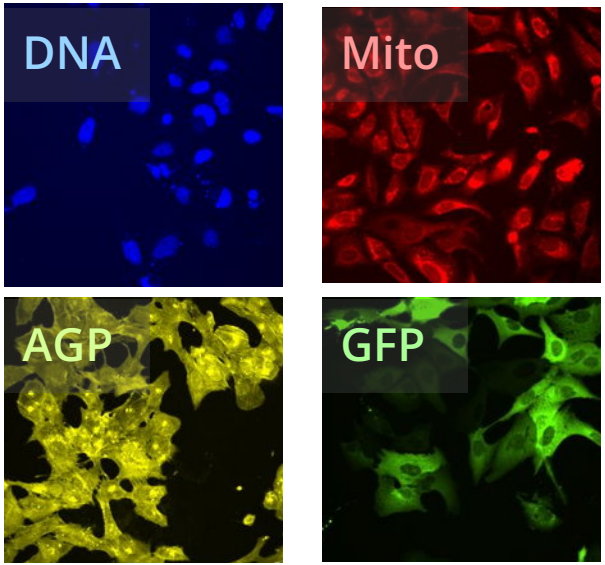
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The Variant Painting Assay (derived from Cell Painting) can profile many phenotypes of genetic variants simultaneously



Standard Cell Painting

(general cell structures)



Variant Painting

(Morphology + Localization)

Different imaging channels and features were used to capture alterations in protein localization and cell morphology

GFP: Protein localization

DNA, Mito, AGP: Cell morphological profiles

Lacoste and Haghighi, et al. Cell 2024

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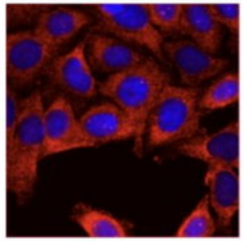
Transforming Variant Painting images into human-designed numerical features by the CellProfiler software

1 - Sample prep



100's of 384-well plates

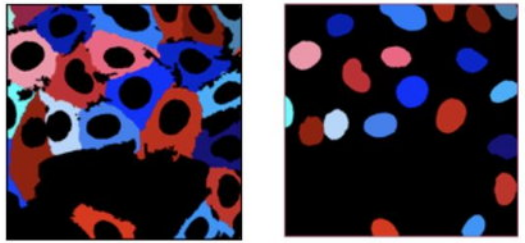
2 - Microscopy



CellProfiler™ cell image analysis software

1000's of images/plate; 4 channels/image

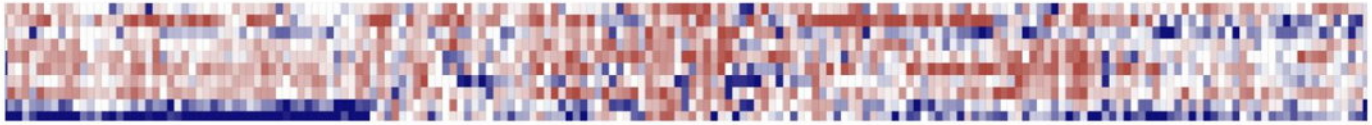
3 - Image segmentation

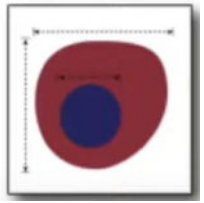


~500 cells/ image

4 - Measure everything

Counts, Shapes, Sizes, Intensities, Textures, Correlations, Learned Features

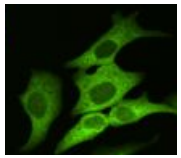




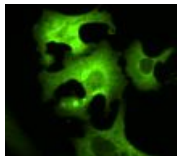
~2000 features / cell

An automated machine learning workflow detects variant cells with protein mislocalization

Imaging wells

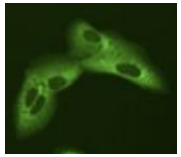


Control Allele

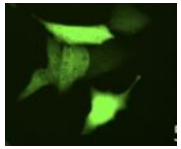


Control Allele

Well position effect



Gene Ref Allele



Gene Var Allele

True mislocalization

Single-cell profiles for reference and variant cells



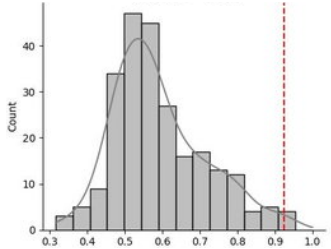
Train Test

Can we predict Ref vs Var by a machine learning classifier?

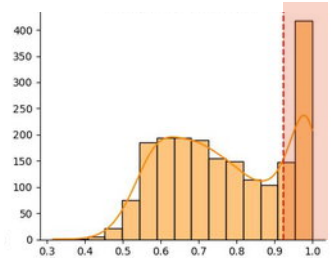


Quantify how well we can predict: Model's AUROC

Distribution of AUROC scores between Ctrl-Ctrl and Ref-Var wells



Null distribution for technical variations caused by well-position effects
95th percentile cutoff for calling hits



Hits for protein mislocalization or altered morphology

Imaging Phenotype Score: is there a significant protein mislocalization between the Variant and Reference alleles?

Ref: healthy-status gene
Var: single amino acid mutation from Ref

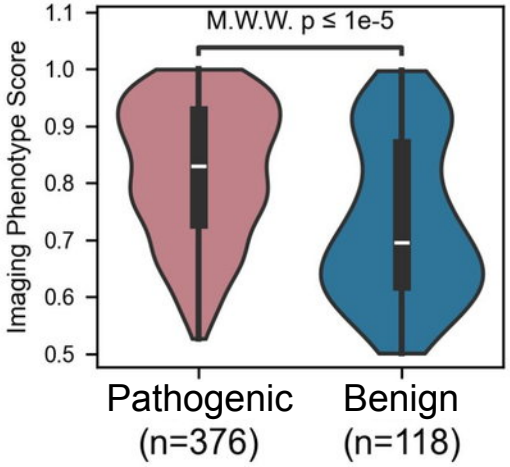


Jess Ewald

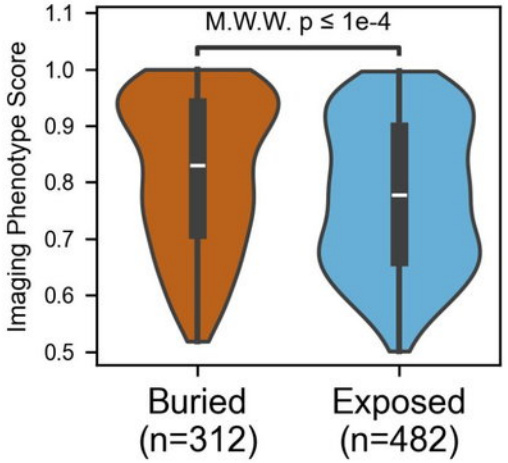
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Protein mislocalization is closely related to a variant's clinical impact and its structural information

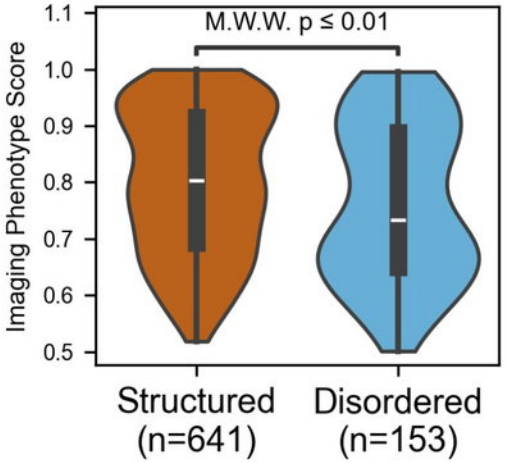
Clinical implications



Global protein structure



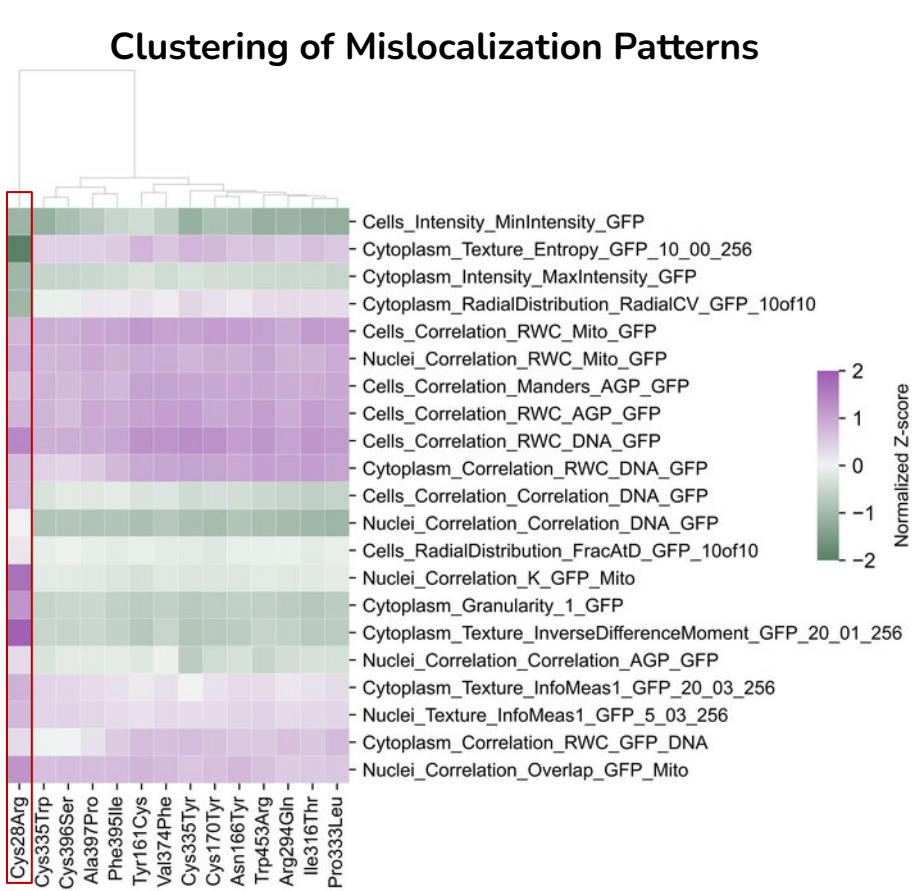
Local protein structure

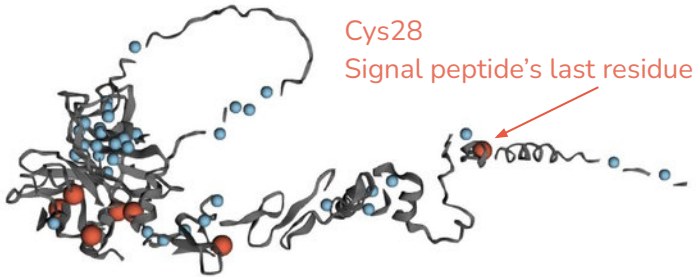


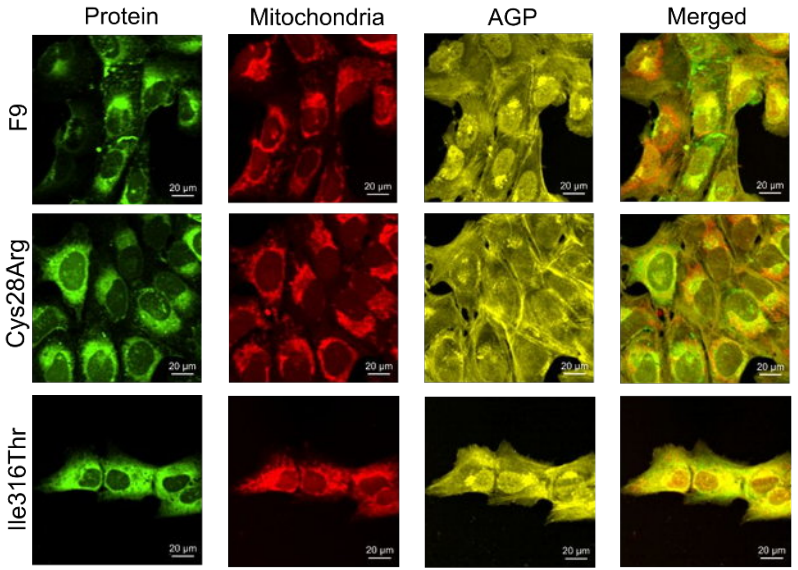
M Tejura, ..., **R Shen**, et al., 2026, *bioRxiv*; F Laval & **R Shen**, et al., 2026, Manuscript in Preparation

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Variants in specific locations of a protein would cause a distinct mislocalization signature





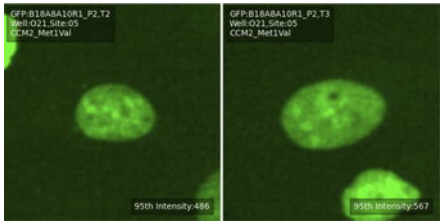


M Tejura, ..., **R Shen**, et al., 2026, *bioRxiv*

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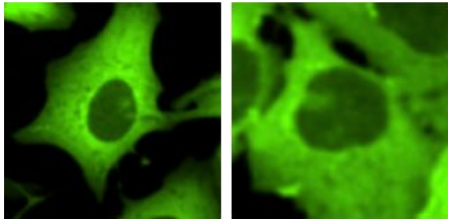
Sometimes variants fall into different categories of functional impacts within the ***same*** gene

1) CCM2 Met1Val



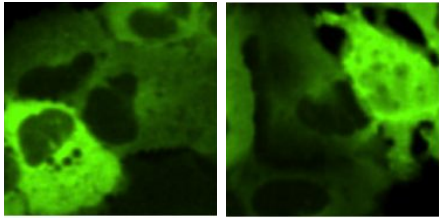
Mislocalized to nuclei only

CCM2 Reference



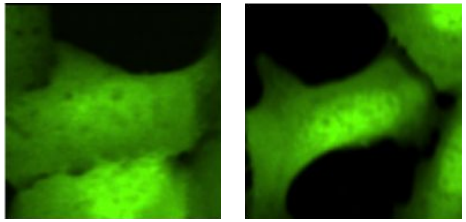
Cytoplasmic localization

4) CCM2 Val53Ile



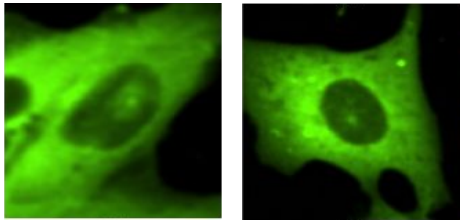
Peripheral redistribution of the protein in the cytoplasm toward the edge of the cell.

2) Ile432Thr Ile425Thr



Mislocalized to nuclei while still expressed in cytoplasm

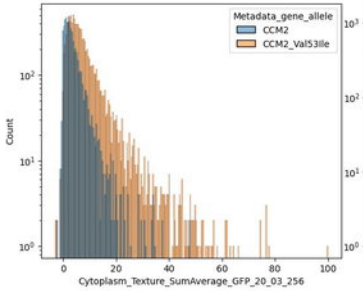
3) Asp376His Ile372Leu



More subtle distribution shifts measured by multiple features

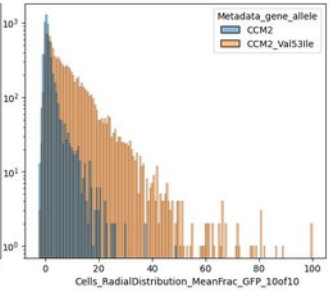
Smoothness of protein distribution
Colocalization with Mito

Texture



Measuring smoothness/roughness

MinIntensityEdge

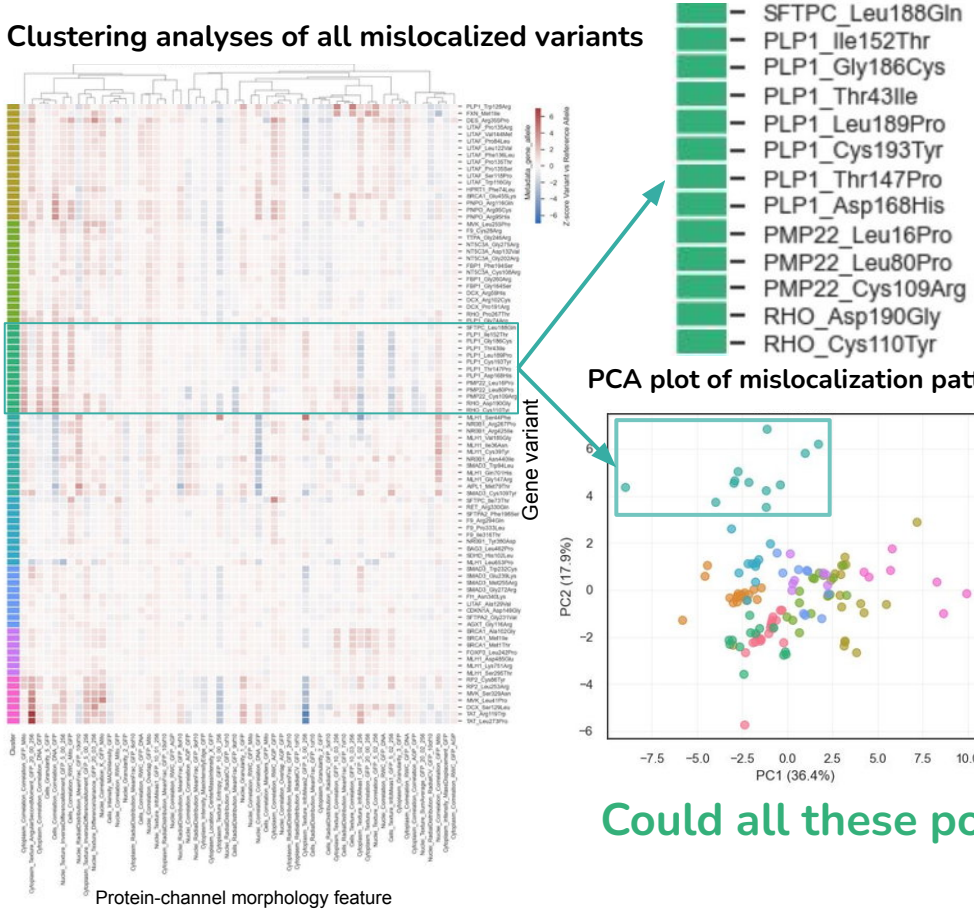


Measuring intensity in the peripheral cytoplasm

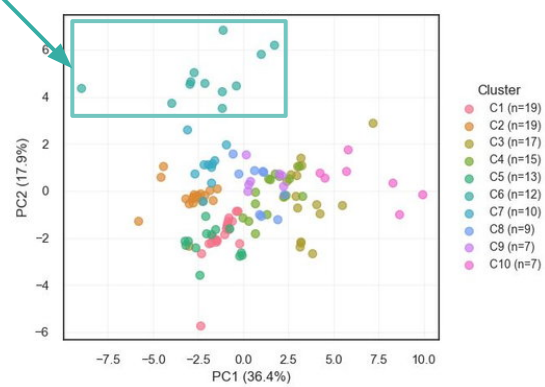
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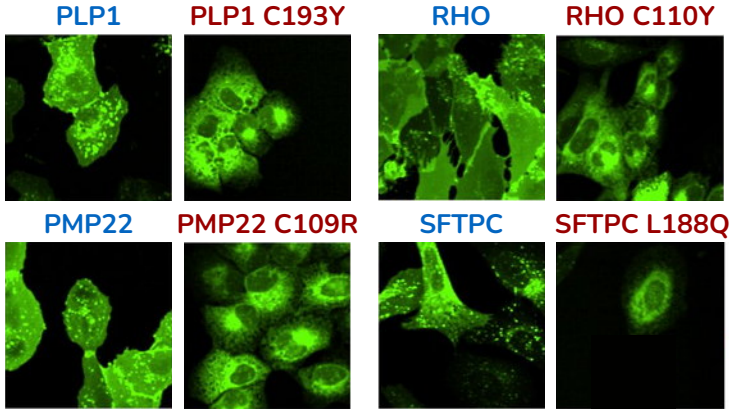
Shared mislocalization patterns were observed across different genes, cell types, and disease modules

Clustering analyses of all mislocalized variants



PCA plot of mislocalization patterns



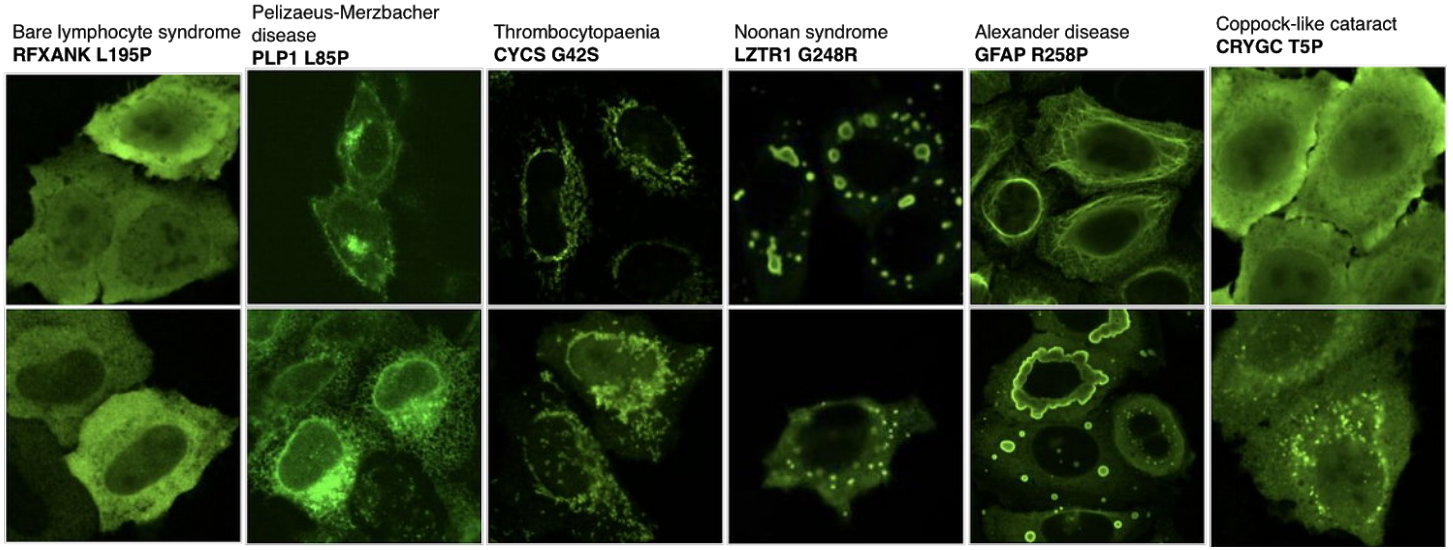


Gene	Protein	Cell Type	Disease
PLP1	Proteolipid protein 1	CNS oligodendrocytes	Hypomyelinating leukodystrophies
PMP22	Peripheral myelin protein 22	Schwann cells	Peripheral neuropathies
RHO	Rhodopsin	Rod photoreceptors	Retinal degenerative disorders
SFTPC	Surfactant protein C	Alveolar type II epithelial cells	Interstitial lung disease

Could all these potentially be reversed with the same drug?

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Now that we found phenotypes, can we find drugs to treat them?



Lacoste, et al. Cell 2024

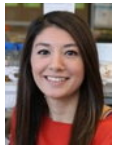
4,500 human genetic variants tested → 250 disease phenotypes discovered

NIH IGVF Consortium: test 30,000 new variants

VISTA Consortium: screen ~300 drugs x ~100 diseases



Mikko Taipale, U Toronto



Jessica Lacoste

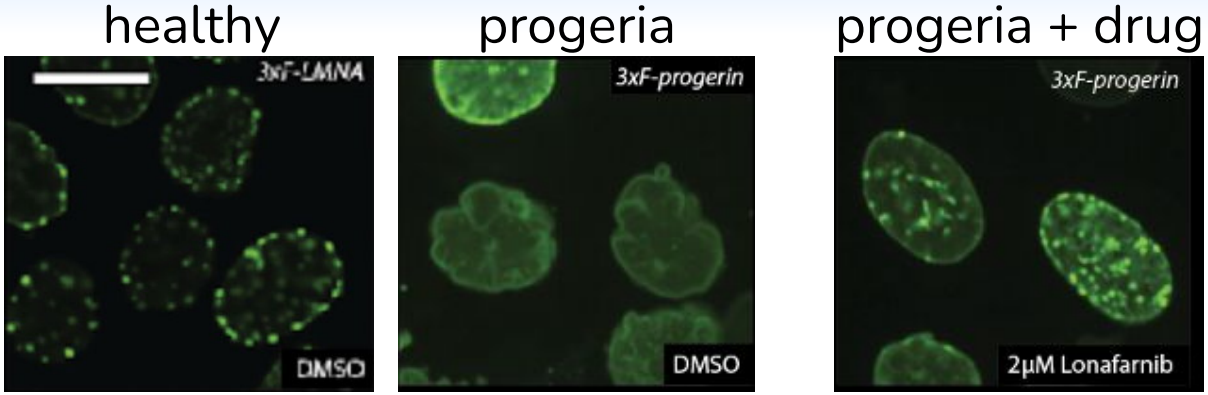


Marzieh Haghighi

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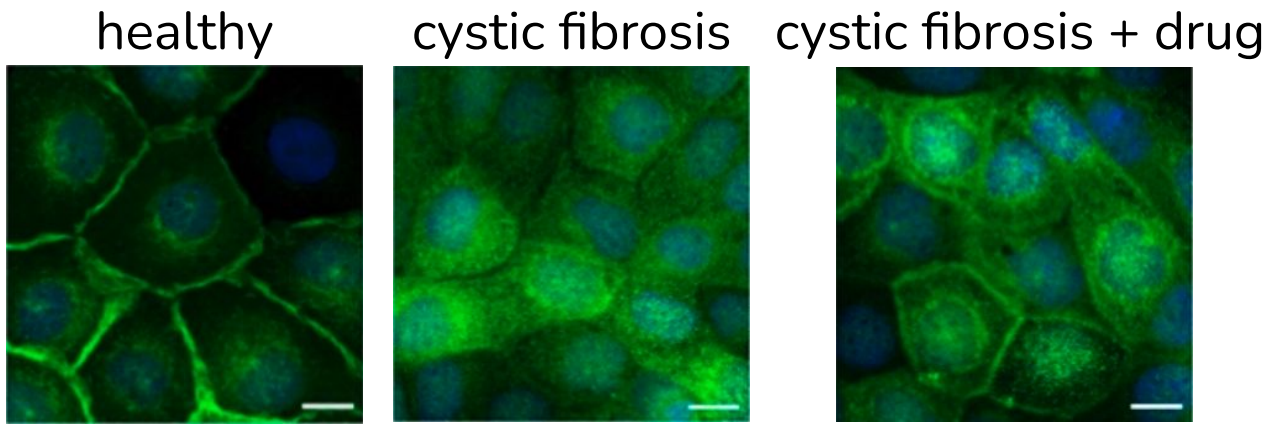
Some successful drugs reverse disease mislocalizations

Lonafarnib relocalizes LMNA in progeria syndrome (HGPS)



Lacoste, et al. unpublished

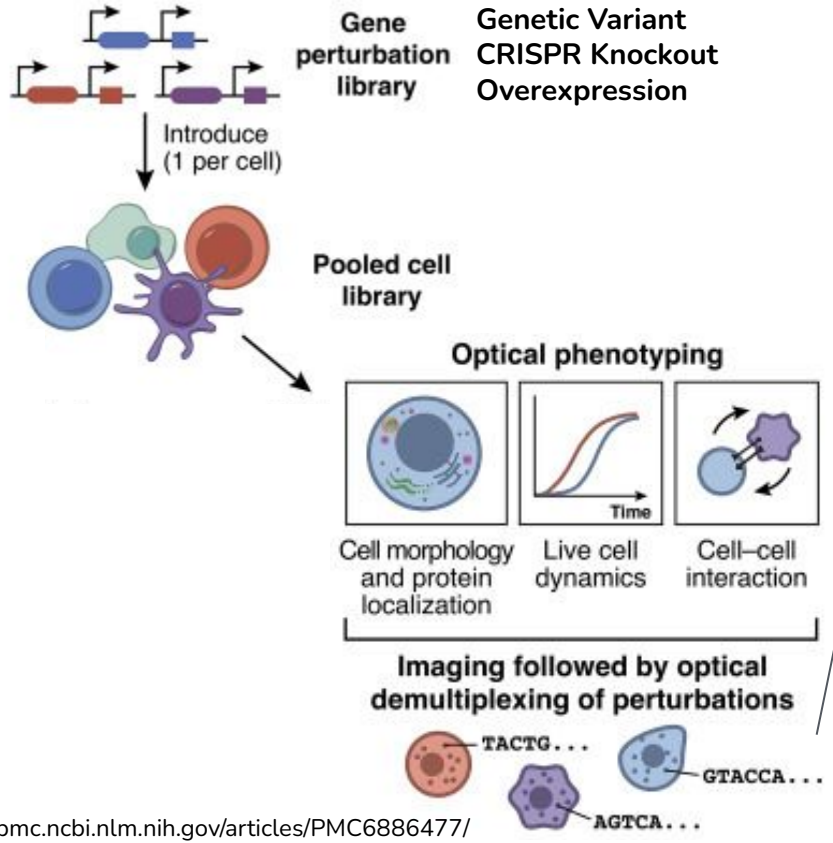
Lumacaftor relocalizes CFTR in cystic fibrosis



Capurro, et al. IJ Mol Sci 2021

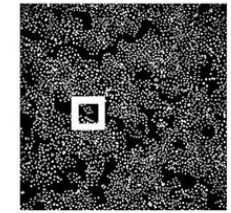
Optical Pooled Screens scale up imaging of complex cellular phenotypes

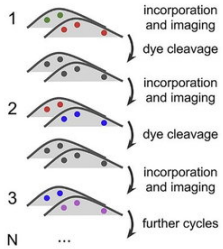
Imaging both the phenotype and the perturbation barcode



https://pmc.ncbi.nlm.nih.gov/articles/PMC6886477/

Reading Perturbation Barcodes by In Situ Sequencing
using fluorescent signals and a microscope instead of a sequencer


10X objective; 3,552 cells

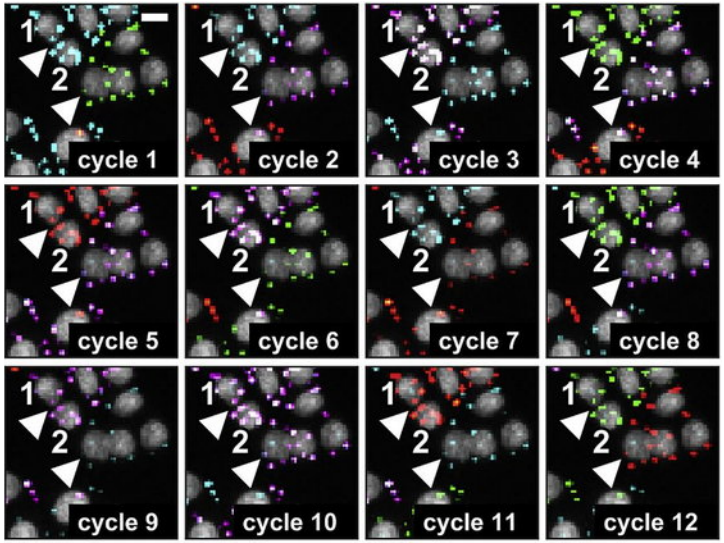


Multiple cycles of fluorescent nucleotide incorporation, imaging, and dye cleavage

barcode 1
CCAGTACGAATG

barcode 2
GACAAGTACACT

1 — 12
sequencing cycle



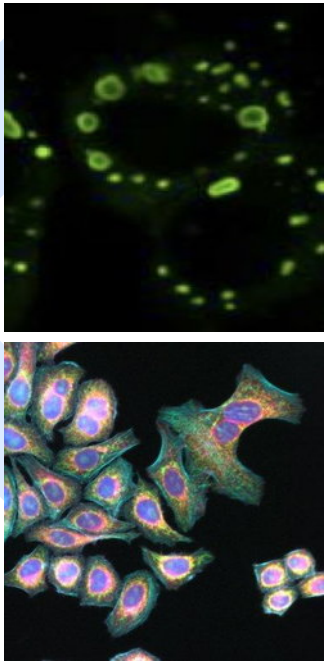
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Scaling up drug discovery with Optical Pooled Screens



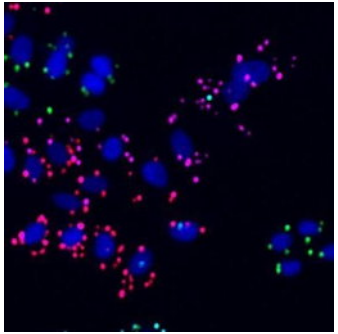
Each well =
1 compound +
200 reference genes
200 disease variants

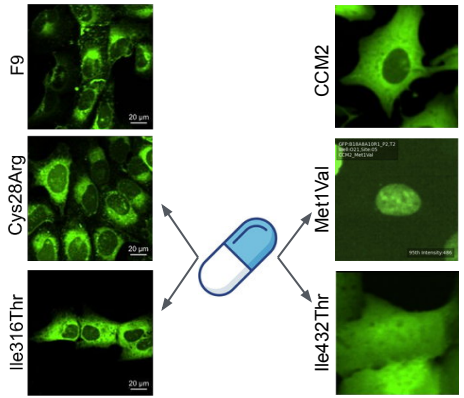
Localization & morphology



+

in situ barcoding



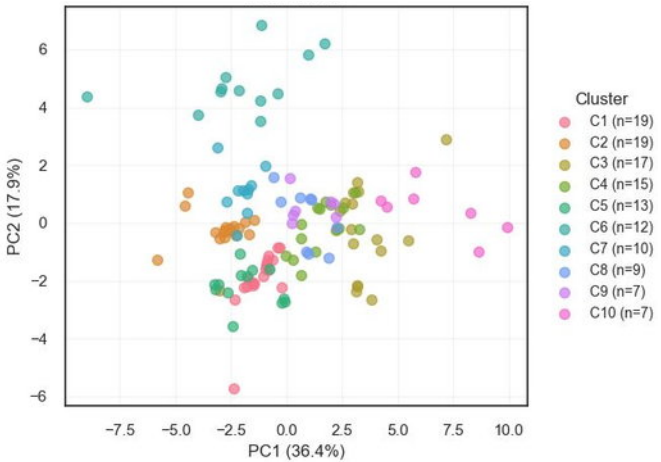


Project in progress, based on Pooled Optical Profiling: Feldman, et al. Cell 2019

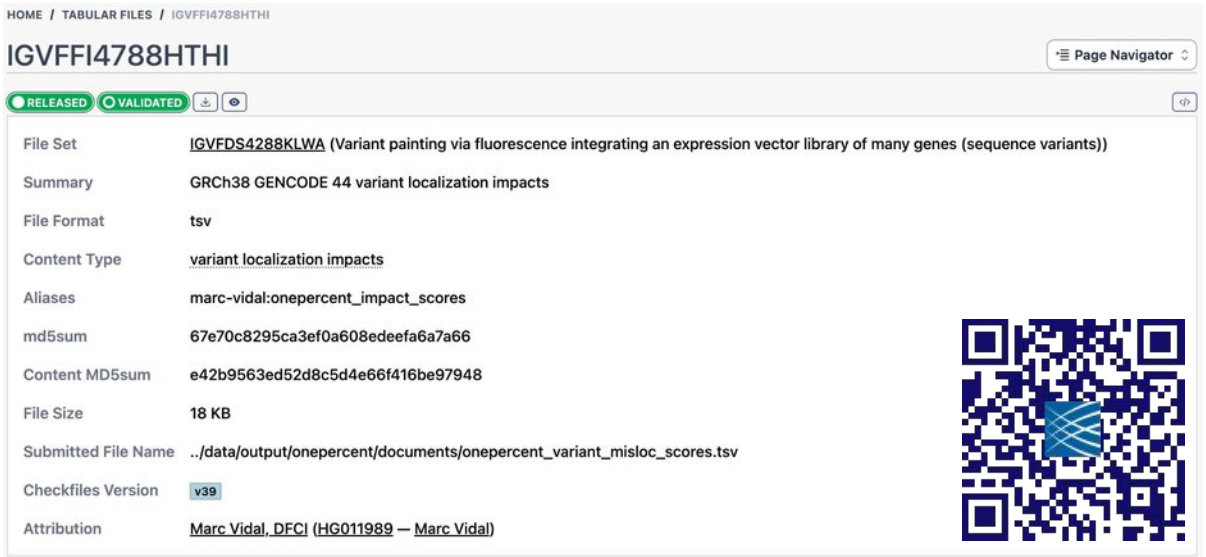
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Most NIH-funded studies share their data publicly and you can easily access them for your own research

Publicly available functional scores per variant on IGVF:
http://broad.io/VariantPainting



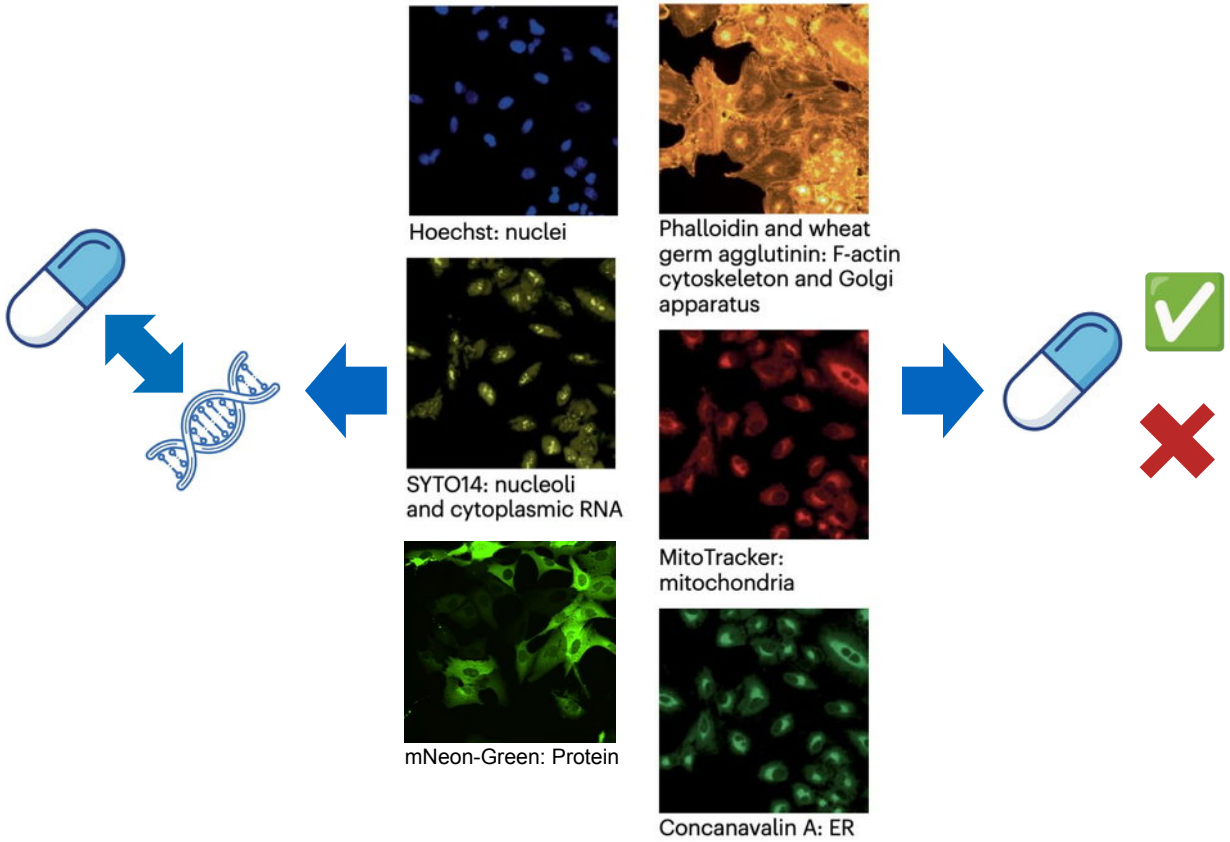
A Cell Map of Variants' Protein Features
Normalized to Respective Reference Alleles



M Tejura, ..., **R Shen**, et al., 2026, *bioRxiv*; IGVF Consortium

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Questions so far?



Can we predict drug and chemical toxicity from images?

30% of Phase 1 and 25% of Phase 2 clinical trials fail due to tox missed by animal models

300,000 commercial compounds have never been assessed for safety!

FDA NEWS RELEASE

FDA Announces Plan to Phase Out Animal Testing Requirement for Monoclonal Antibodies and Other Drugs

For Immediate Release: April 10, 2025

THE NIH DIRECTOR

The NIH Director

- Statements
- Testimonies
- Advisory Groups
- Previous NIH Directors

February 1, 2024

Statement on catalyzing the development of novel alternative methods

NIH National Institutes of Health
Office of Strategic Coordination–The Common Fund

Search Common Fund

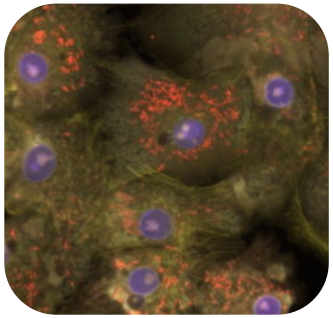
Our Programs | Research Funding | News & Media | New Program Planning | Evaluation & Assessment

Complement Animal Research In Experimentation (Complement-ARIE) Program

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US Environmental Protection Agency uses Cell Painting for compound toxicity testing

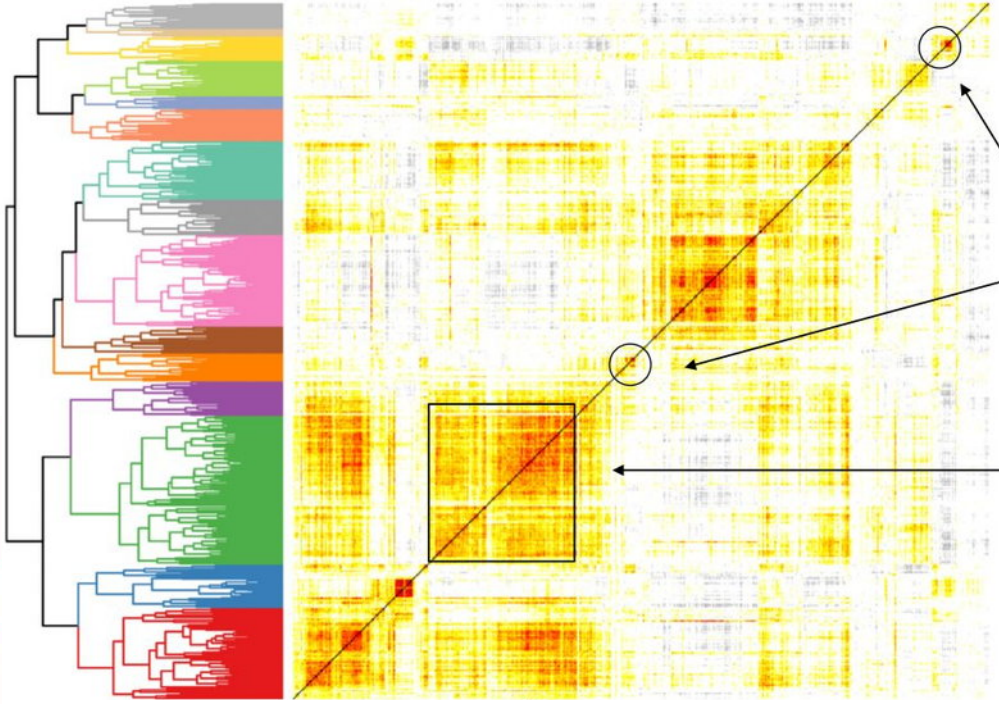
Clustering based on morphological similarity reveals similar mechanisms



Cyproconazole	Tebuconazole
Fenbuconazole	Tetraconazole
Hexaconazole	Triadimenol
Imazalil	Triadimefon
Paclobutrazol	Triticonazole

Diniconazole

Difenoconazole
Ketoconazole
Prochloraz
Propiconazole



Slide courtesy of Johanna Nyffeler and Joshua Harrill

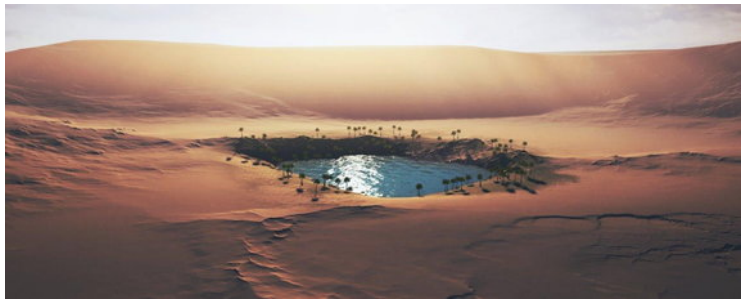
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Consortia can tackle problems too big for any one lab or company: predicting toxicity of thousands of compounds

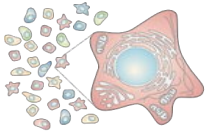
OASIS

Omics for Assessing Signatures for Integrated Safety Consortium

"Successful drug discovery is like finding oases of safety and efficacy in chemical and biological deserts." – *Scannell, Nat Rev Drug Discovery 2022*

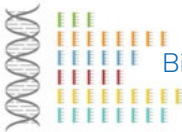


Cell Painting (images)



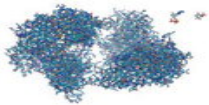
Broad

Transcriptomics (mRNA)



BioSpyder

nELISA (protein)



Nomic bio

Predict compound toxicity in human

Rouquié, et al. Tox Sci 2025

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OASIS Consortium for toxicity assessment

OASIS
Omics for Assessing Signatures for Integrated Safety Consortium

Public and private compounds with associated in vivo and/or human data

- ❑ 200 donated from industry partners
- ❑ 1,400 from public domain

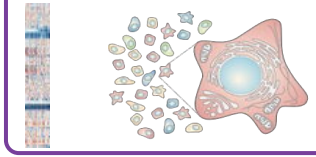


Cell model systems *(of increasing complexity, physiological relevance, and cost)*

- ❑ U2OS human osteosarcoma immortal cell line
- ❑ HepaRG human primary hepatocyte-like cell line
- ❑ Primary human hepatocytes
- ❑ Primary rat hepatocytes
- ❑ Liver organoids, Liver-on-a-chip

Multi-omic profiling assays

Cell Painting (images)



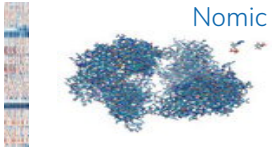
Transcriptomics (mRNA)

BioSpyder



nELISA (protein)

Nomic bio



Data analysis

- ❑ Translating *in vitro* to *in vivo* concentrations (qIVIVE)
- ❑ Distinguishing & interpreting mechanisms
- ❑ Integrating evidence across multiple data types/cell models

Publicly accessible data
Published methodologies

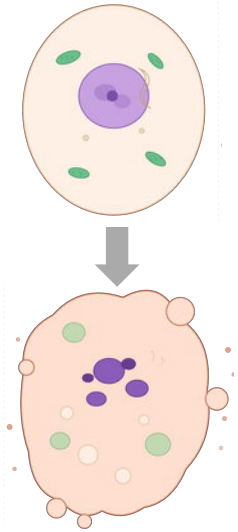
- ❑ Community of Practice



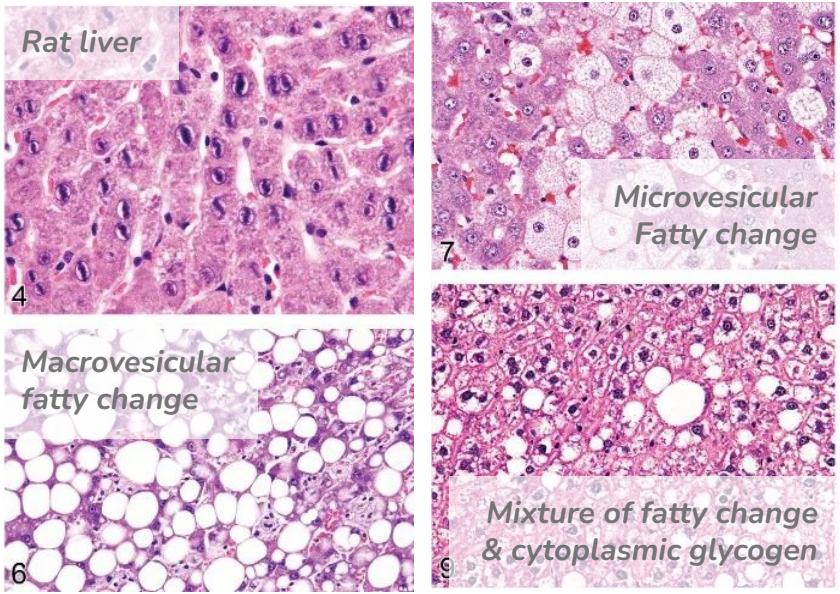
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Compounds are not just toxic or non-toxic: there are multiple types of toxicities such as cytotoxicity and many specific liver toxicities

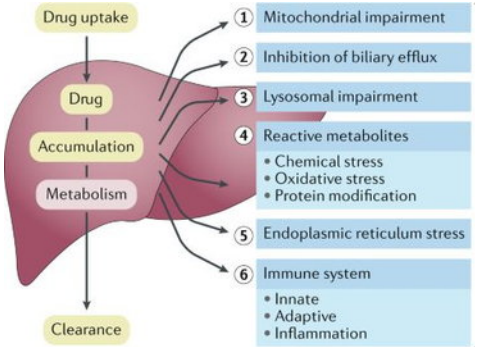
Cell cytotoxicity



Rat liver toxicity



Human drug-induced liver injury (DILI)



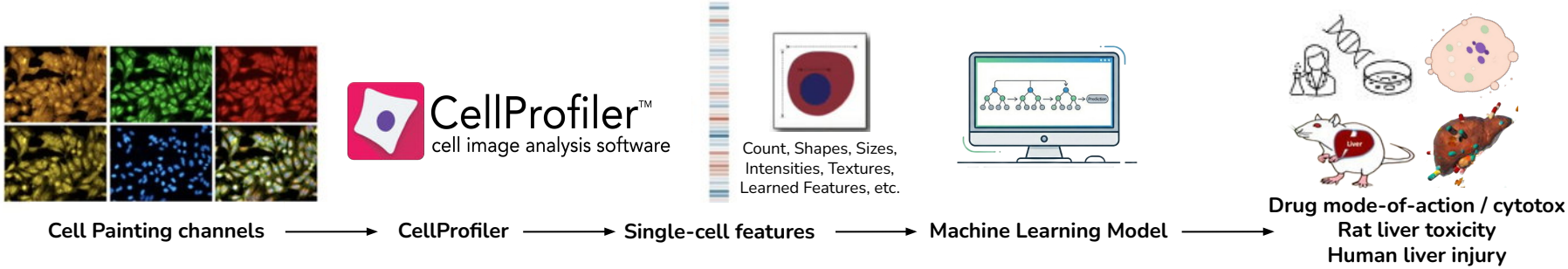
Diverse clinical presentations of DILI

- Acute fatty liver with lactic acidosis
- Acute hepatic necrosis
- Acute liver failure
- Acute viral hepatitis-like liver injury
- Autoimmune-like hepatitis
- Bland cholestasis
- Cholestatic hepatitis
- Cirrhosis
- Immuno-allergic hepatitis
- Nodular regeneration
- Nonalcoholic fatty liver
- Sinusoidal obstruction syndrome
- Vanishing bile duct syndrome

- ❑ **Goal:** Predict a specific toxicity event using Cell Painting data
- ❑ **Data curation:** careful data curation with expert knowledge is essential to downstream prediction tasks

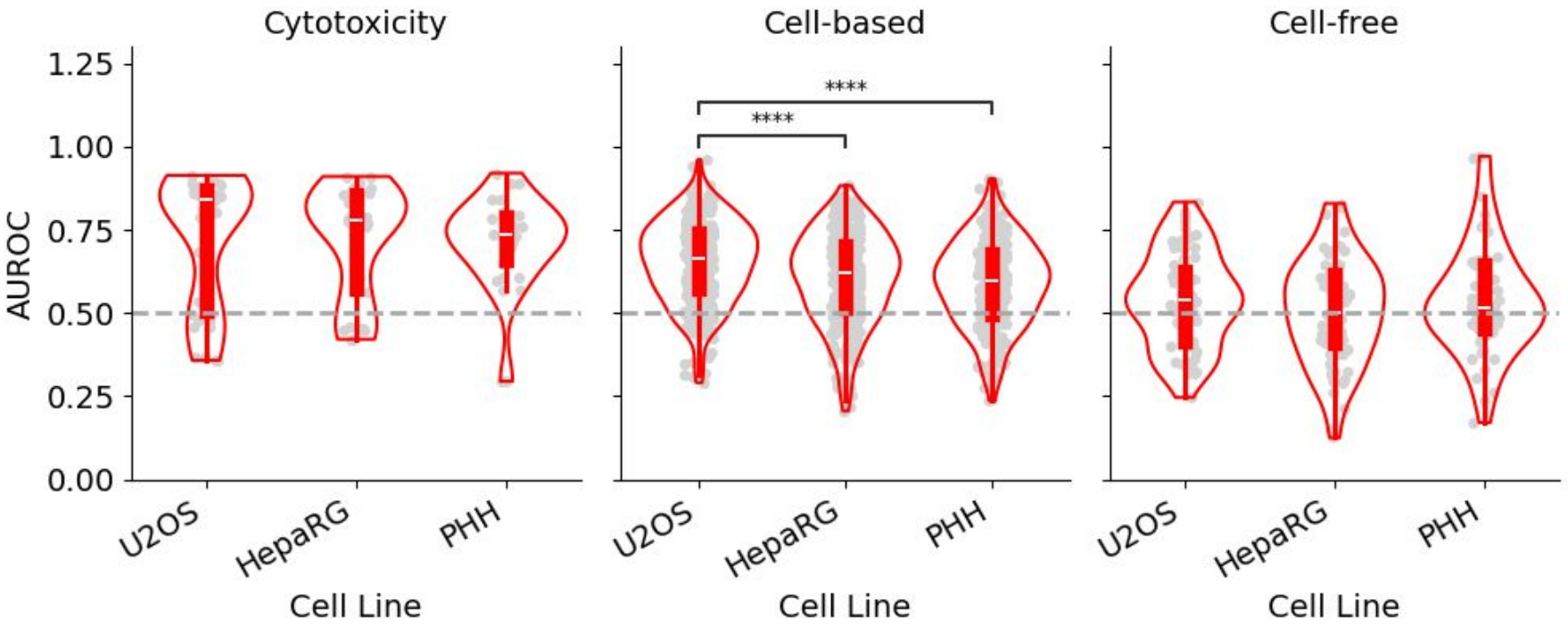
https://focusontoxpath.com/proliferative-and-nonproliferative-lesions-of-the-rat-and-mouse-hepatobiliary-system/#
https://www.nature.com/articles/s41573-019-0048-x

Can simple cell cultures predict and explain bioactivity and liver toxicity in humans and rats?



- ❑ **Predict compound's in vitro cytotoxicity and mode-of-action (MOA: what the drug does to the body at the molecular level)**
- ❑ **Predict and protect against hepatotoxicity endpoints in humans and rats**
- ❑ **How much does it matter how liver-relevant the cell type is?**
  - ❑ Human U2OS vs. HepaRG vs. primary human hepatocyte (PHH)
- ❑ **How much does computational method matter?**
  - ❑ Supervised machine learning for predicting how likely a specific toxicity event would happen
  - ❑ Feature sets: 1. CellProfiler morphological features | 2. Cell count | 3. Random baseline

Different cell lines have similar performances in predicting cytotoxicity but different performances in predicting complex drug effects



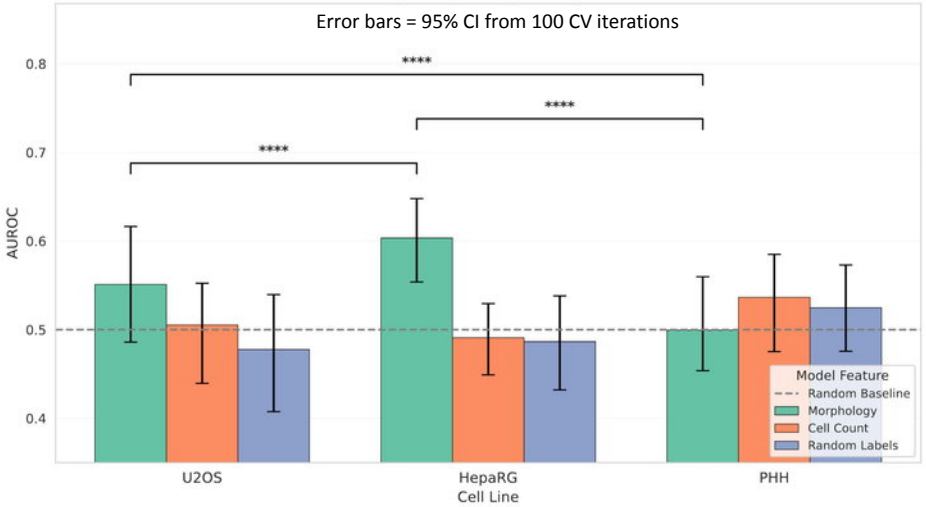
In vitro cytotoxicity **Mode-of-action** **Target binding**

J Ewald, et al., 2025, *bioRxiv*

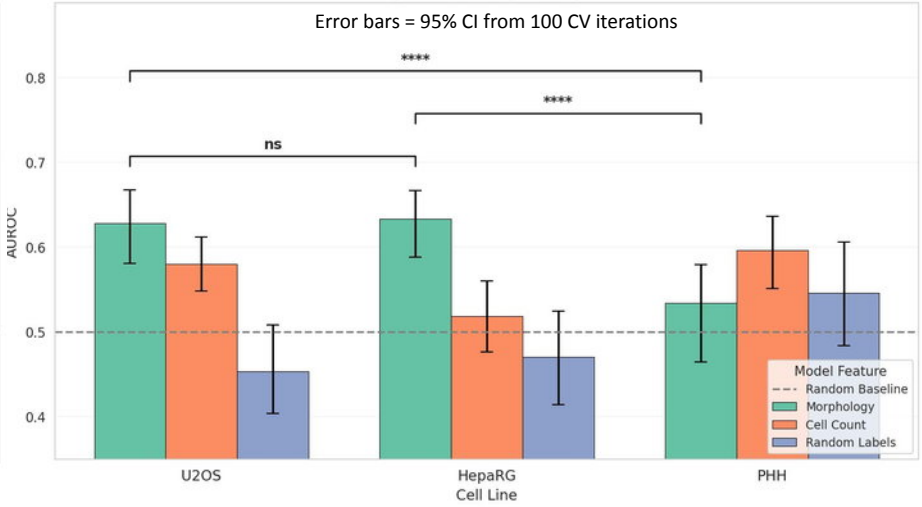
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Different cell lines have different performances in predicting rat liver toxicity and human drug-induced liver injury

HepaRG best predicted rat hepatic toxicity, followed by U2OS, while primary human hepatocyte showed no signal



HepaRG and U2OS significantly outperform primary human hepatocyte for human DILI prediction



Why primary human hepatocyte might underperform:

- ❏ Rapid dedifferentiation blurs morphological signals
- ❏ Donor-to-donor variability reduces reproducibility
- ❏ Erases the consistent stress-response signatures

Why HepaRG excels at rat hepatotoxicity:

- ❏ Stable clonal cultures with robust hepatic metabolism
- ❏ Liver-specialized pathways translate rat toxic mechanisms into detectable cellular phenotypes

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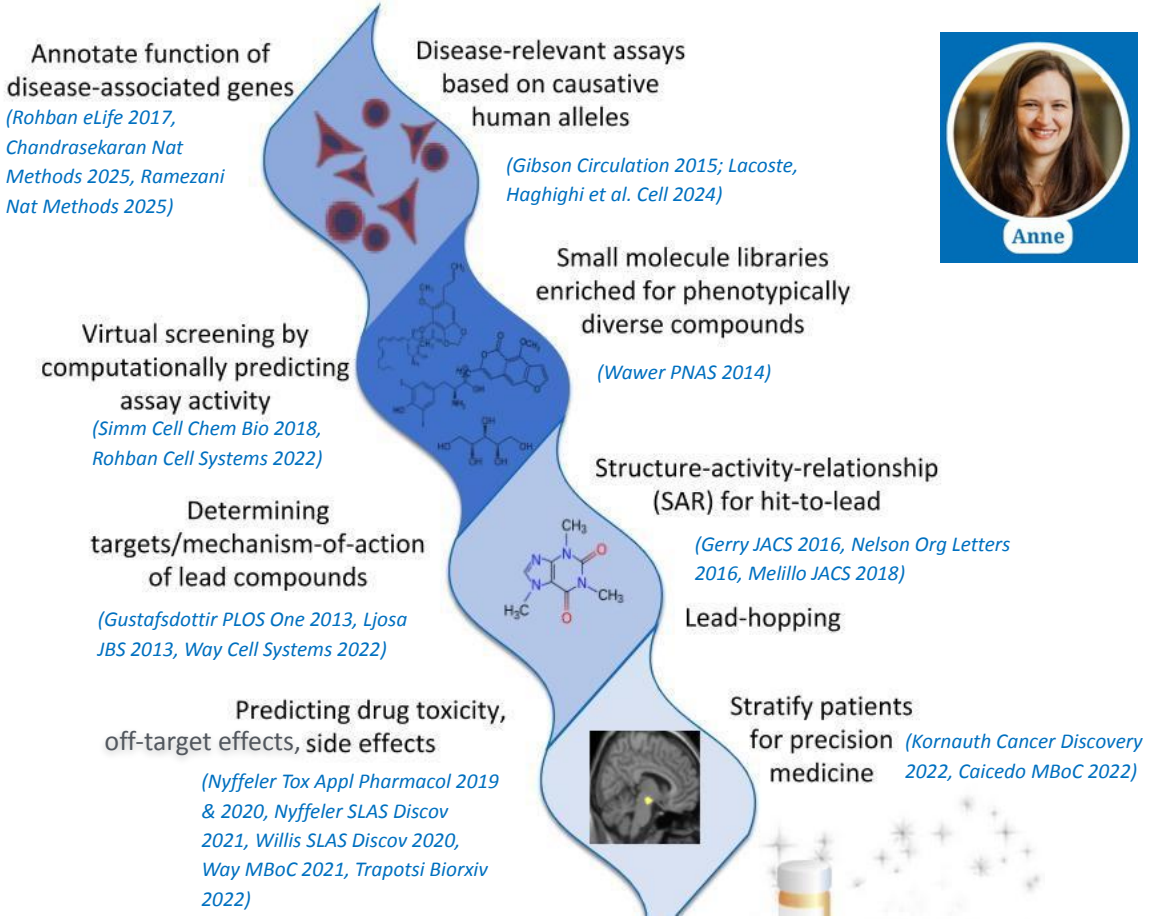
Cell Painting: a decade of discovery and innovation in cellular imaging

Srijit Seal, Maria-Anna Trapotsi, Ola Spjuth, Shantanu Singh, Jordi Carreras-Puigvert, Nigel Greene, Andreas Bender & Anne E. Carpenter

Nature Methods **22**, 254–268 (2025) | Cite this article

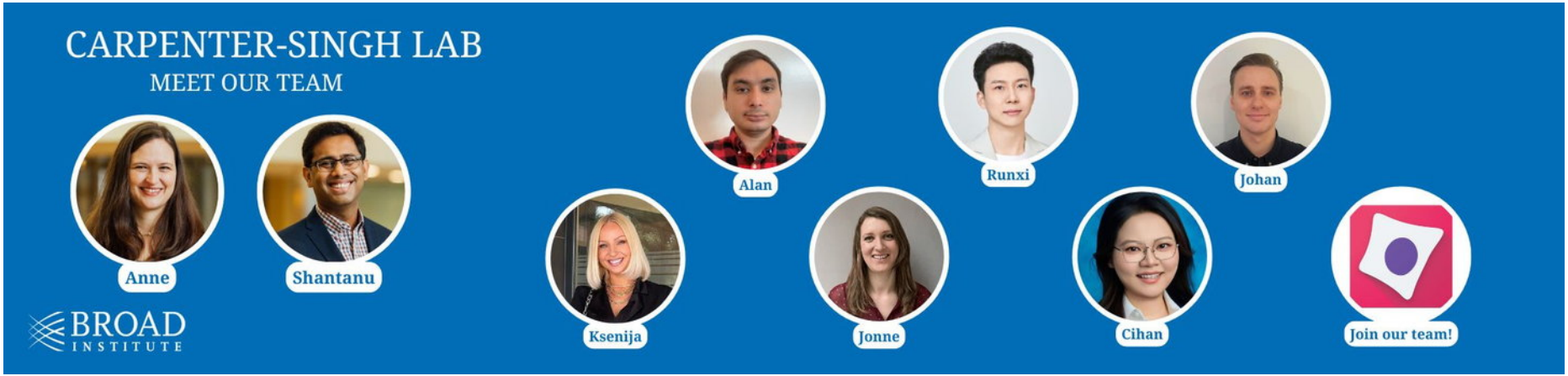
Can we speed up individual steps using automation, clever biotech, AI agents?

Can we **surpass human capabilities**?



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Chloe Reno
Fritz Roth
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